LOW-VOID INDIUM DIE ATTACH FOR PHOTONIC PACKAGING

VACUUM REFLOW PROCESS DEVELOPMENT, VOID REDUCTION AND BOND QUALIFICATION

TOOLS & METHODS:

Plasma surface preparation (oxygen and argon); programmable vacuum reflow oven with controlled vacuum, gas flow and temperature profile; custom alignment and pressure fixtures; indium preforms with formic acid process gas and nitrogen; X-ray void imaging with image-analysis quantification; destructive die shear and wire bond pull testing; automated large-format wire bonding.

EXECUTIVE SUMMARY:

Photonic integrated circuits and low-noise amplifier dice in quantum computing hardware cannot tolerate the peak temperatures of conventional solder attach, yet they depend on void-free bonds for both mechanical strength and heat transfer. This paper describes a vacuum reflow process built on indium and formic acid vapor that bonds these die at low temperature, with a peak of 185 °C.

Measured by X-ray, void area fell from **25.29%** with a manual die attach to **4.86%** after profile optimization, roughly a five-fold reduction. The process is a fixture-controlled, repeatable sequence, qualified with X-ray void analysis, destructive die shear and wire bond pull testing.

KEY RESULTS:

- Void area cut from **25.29%** (manual attach) to **4.86%** (optimized vacuum reflow)
- Profile optimization alone reduced voiding from 15.94% to 4.86%, a **69% relative reduction**
- Peak process temperature of **185 °C**, with indium melting at 157 °C
- Flux-free bonding: formic acid vapor strips oxides in place of traditional flux
- Qualified by X-ray void inspection, die shear testing and wire bond pull testing

INTRODUCTION:

Photonic approaches to quantum computing place unusual demands on packaging. A single package can combine a photonic integrated circuit (PIC), low-noise amplifier (LNA) dice and a range of miniature components, and the structures inside the PIC are sensitive to heat. The attach process therefore has to work at low temperature while still delivering a strong, thermally conductive bond.

Two properties drive the requirements. Voids weaken a bond and interrupt the heat path, so low void content matters for both strength and thermal performance. Low-temperature processing protects the sensitive device structures. **Indium** was selected as the bonding material for its 157 °C melting point and high ductility, and **formic acid** was selected as the process gas because it performs well below 200 °C.

Process development covered the full assembly flow: surface preparation, aligned placement in custom fixtures, vacuum reflow, inspection and wire bonding (FIG 1).

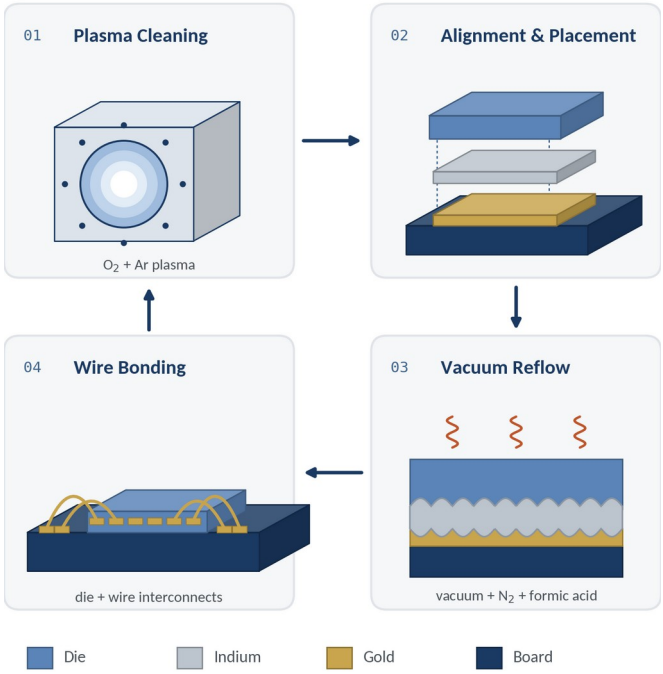


FIG 1: Process flow for vacuum reflow eutectic die bonding, from plasma cleaning through wire bonding.

TEST SETUP:

Plasma cleaning. The substrate, die and indium are cleaned before reflow to minimize moisture, remove oxides and eliminate organics. A controlled oxygen and argon plasma strips contaminants from the surfaces. Clean surfaces are the main lever for a bond with minimal voiding, and they protect bond strength under shear testing.

Alignment and fixturing. Custom fixtures position the indium and die and apply controlled pressure during bonding. The aligned assembly then goes into the vacuum reflow oven, where a programmed profile sets vacuum level, gas flow and temperature (FIG 2).

Vacuum reflow sequence. The profile runs in four stages:

- Draw vacuum, purge with nitrogen, then draw a second vacuum so no oxygen remains in the chamber.
- Fill with formic acid and heat to just below the indium liquidus, then hold so the formic acid can clean the surfaces.
- Heat above the liquidus so the indium is fully molten and pull vacuum again to remove trapped bubbles and mitigate voids.
- Hold at 185 °C for several minutes, then cool with nitrogen and water.

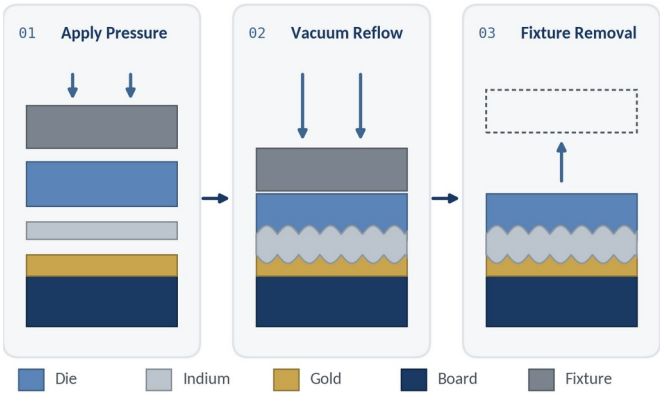


FIG 2: Bond stack during pressure application, vacuum reflow and fixture removal.

Inspection. After fixture removal, each part goes through X-ray void inspection and destructive die shear testing against the minimum requirements of the quality protocol. X-ray locates voided regions and die shear measures the bond strength achieved.

Wire bonding. A large-format automated wire bonder forms the electrical interconnects between the substrate and the LNA components. Between 3 and 5% of wire bonds are destructively pull tested to give statistical validation of mechanical robustness.

Void quantification. An image-analysis script segments voids by grayscale contrast and reports voided area as a percentage of the die bond region, so every sample is compared on the same basis.

RESULTS:

A manual die attach using the same die and indium provided the baseline. Die shear testing ran iteratively through development to refine the bonding parameters, and void analysis after reflow continues to be used to confirm process performance.

Condition	Void area	vs. 25% limit
Manual die attach	25.29%	Just above
Vacuum reflow, unoptimized	15.94%	Within
Vacuum reflow, optimized	4.86%	Well within

TABLE 1: Void area by X-ray image analysis. The 25% limit is the IPC-A-610 criterion for voiding per joint.

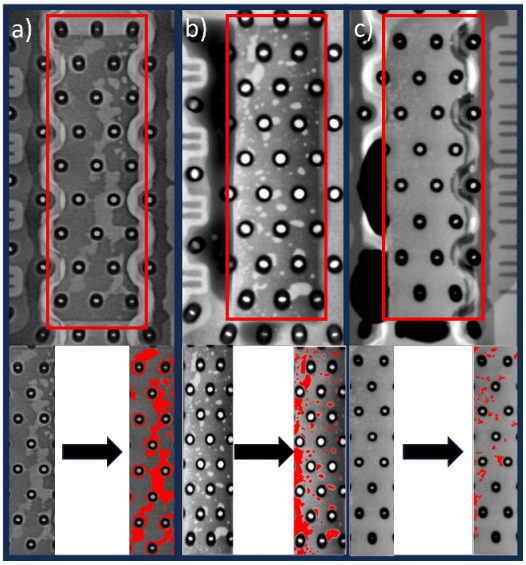


FIG 3: X-ray scans (top) and analyzed regions (bottom) for (a) manual die attach, 25.29% void; (b) unoptimized vacuum reflow, 15.94%; (c) optimized vacuum reflow, 4.86%.

The manual attach landed just above the 25% void limit. Likely causes are trapped air, uneven material application and outgassing without a controlled pressure or environment.

Vacuum reflow before profile optimization reached 15.94%. Vacuum alone does not guarantee low voiding: poorly controlled conditions still leave significant voids. After optimization, voiding fell to **4.86%**, about five times lower than the manual process and 69% lower than the unoptimized profile. Large voids were eliminated as well as the overall void percentage.

APPLICATIONS:

- Die attach for thermally sensitive photonic integrated circuits and low-noise amplifier dice
- Flux-free, low-temperature bonding where residue or high process temperature would damage the device
- Packages where bond strength and heat transfer both depend on low void content
- A repeatable, scalable replacement for manual die attach in next-generation photonics packaging

The combination of plasma preparation, fixtured alignment and profiled vacuum reflow gives a process that can be characterized, tuned and qualified with standard inspection methods.

DISCUSS THIS WORK:

This paper summarizes one project. If a similar die attach, packaging or void-reduction problem appears in your program, I am happy to walk through the development approach and how it would map onto your hardware. emincherif@gmail.com